

# COMPUTER SCIENCE & IT

## DIGITAL LOGIC



Lecture No: 08

Sequential Circuit



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# Recap of Previous Lecture



- Synch. Counter
- 
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# Topics to be Covered

Synch. Counter cont.



- Design counter 0 – 1 – 2 – 5 – 6-3 using D-FF

MOD no. = 6  
3 ffs

| $Q_2$ | $Q_1$ | $Q_0$ | $Q_2^+$ | $Q_1^+$ | $Q_0^+$ | $D_2$ | $D_1$ | $D_0$ |
|-------|-------|-------|---------|---------|---------|-------|-------|-------|
| 0     | 0     | 0     | 0       | 0       | 1       | 0     | 0     | 1     |
| 0     | 0     | 1     | 0       | 1       | 0       | 0     | 1     | 0     |
| 0     | 1     | 0     | 1       | 0       | 1       | 1     | 0     | 1     |
| 0     | 1     | 1     | 0       | 0       | 0       | 0     | 0     | 0     |
| 1     | 0     | 0     | X       | X       | X       | X     | X     | X     |
| 1     | 0     | 1     | 1       | 1       | 0       | 1     | 1     | 0     |
| 1     | 1     | 0     | 0       | 1       | 1       | 0     | 1     | 1     |
| 1     | 1     | 1     | X       | X       | X       | X     | X     | X     |



$$D_2[Q_2, Q_1, Q_0] = \Sigma(2, 5) + d\Sigma(4, 7)$$

$$D_1[Q_2, Q_1, Q_0] = \Sigma(1, 5, 6) + d\Sigma(4, 7)$$

$$D_0[Q_2, Q_1, Q_0] = \Sigma(0, 2, 6) + d\Sigma(4, 7)$$

$D_2$

|                  | $\overline{Q_1}\overline{Q_0}$ | $\overline{Q_1}Q_0$ | $Q_1Q_0$ | $Q_1\overline{Q_0}$ |
|------------------|--------------------------------|---------------------|----------|---------------------|
| $\overline{Q_2}$ |                                |                     |          | 1                   |
| $Q_2$            | X                              | 1                   | X        |                     |

$$D_2 = Q_2(n)Q_0(n) + \overline{Q_2}Q_1\overline{Q_0}$$

$D_1$

|                  | $\overline{Q_1}\overline{Q_0}$ | $\overline{Q_1}Q_0$ | $Q_1Q_0$ | $Q_1\overline{Q_0}$ |
|------------------|--------------------------------|---------------------|----------|---------------------|
| $\overline{Q_2}$ |                                | 1                   |          |                     |
| $Q_2$            | X                              | 1                   | X        | 1                   |

$$D_1 = Q_2 + \overline{Q_1}Q_0$$

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 0 | 1 | 1 |

$D_0$

|                  | $\overline{Q_1}\overline{Q_0}$ | $\overline{Q_1}Q_0$ | $Q_1Q_0$ | $Q_1\overline{Q_0}$ |
|------------------|--------------------------------|---------------------|----------|---------------------|
| $\overline{Q_2}$ |                                |                     |          | 1                   |
| $Q_2$            | 1                              |                     |          | 1                   |
| $Q_2$            | X                              |                     | X        | 1                   |

$$D_0 = \overline{Q_0}$$

$$Q_0(n+1) = D_0 = \overline{Q_0}(n)$$

| $Q_2$ | $Q_1$ | $Q_0$ |
|-------|-------|-------|
| 0     | 0     | 0     |
| 0     | 0     | 1     |
| 0     | 1     | 0     |
| 1     | 0     | 1     |
| 1     | 1     | 0     |
| 0     | 1     | 1     |



$$f_{Q_1} = f_{CLK}/6$$

$Q_2 \rightarrow$

$$T_{OFF} = 4T_{CLK}, T_{ON} = 2T_{CLK}$$

$$T_{Q_2} = 6T_{CLK}, f_{Q_2} = f_{CLK}/6$$

$Q_0 \rightarrow$

$$T_{OFF} = 1T_{CLK}, T_{ON} = 1T_{CLK}$$

$$T_{Q_0} = 2T_{CLK} \rightarrow f_{Q_0} = f_{CLK}/2$$



# [ Question ]

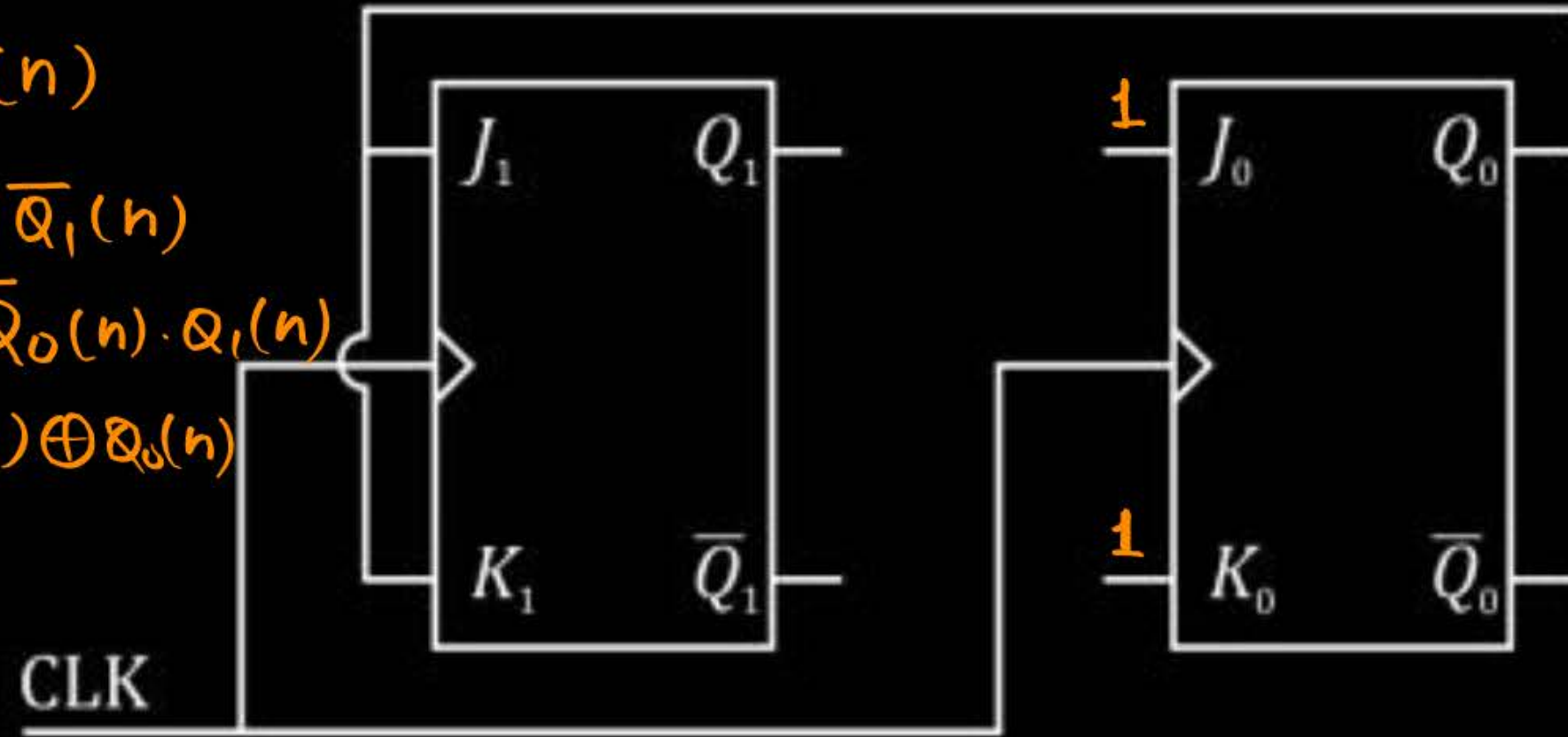
A sequential circuit is as given below :

$$Q_0(n+1) = \bar{Q}_0(n)$$

$$Q_1(n+1) = Q_0(n)\bar{Q}_1(n)$$

$$+ \bar{Q}_0(n) \cdot Q_1(n)$$

$$= Q_1(n) \oplus Q_0(n)$$



|            | $Q_0$ | $Q_1$ |               |
|------------|-------|-------|---------------|
|            | 0     | 0     | $\uparrow$ 00 |
| $\uparrow$ | 1     | 0     |               |
| $\uparrow$ | 0     | 1     |               |
| $\uparrow$ | 1     | 1     |               |

MOD = 4

Initial state of the synchronous counter is given as  $Q_0Q_1 = (00)_2$  what will be states of counter after applying clock pulses.

(a) ✓ 00 – 10 – 01 – 11 – 00

(b) ✗ 00 – 10 – 11 – 01 – 00

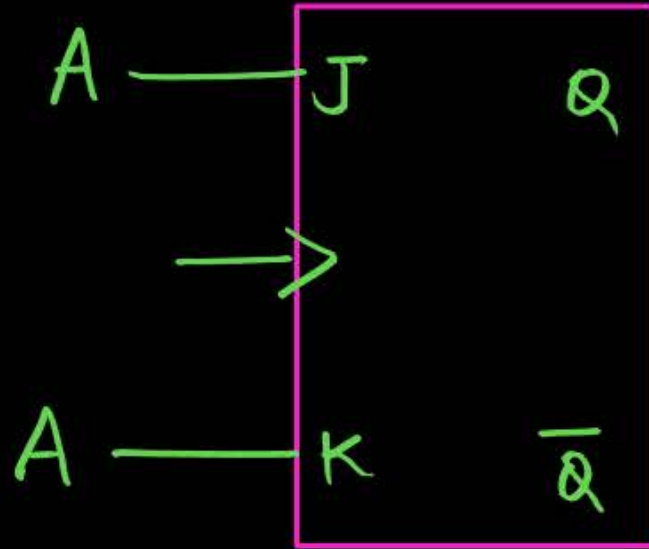
(c) ✗ 00 – 11 – 10 – 01 – 00

(d) ✗ 00 – 01 – 10 – 11 – 00

$$Q_1Q_0 = (00)_2.$$

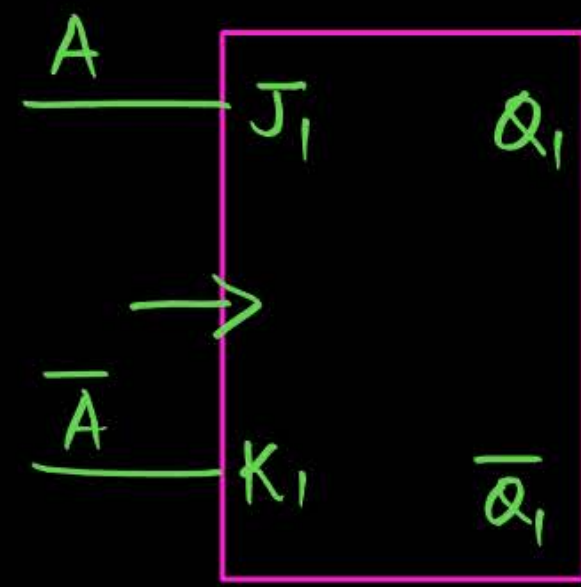
(d) ~~X~~ 00 - 01 - 10 - 00





$$Q(n+1) = A\bar{Q}(n) + \bar{A}Q(n)$$

$$= A \oplus Q(n)$$



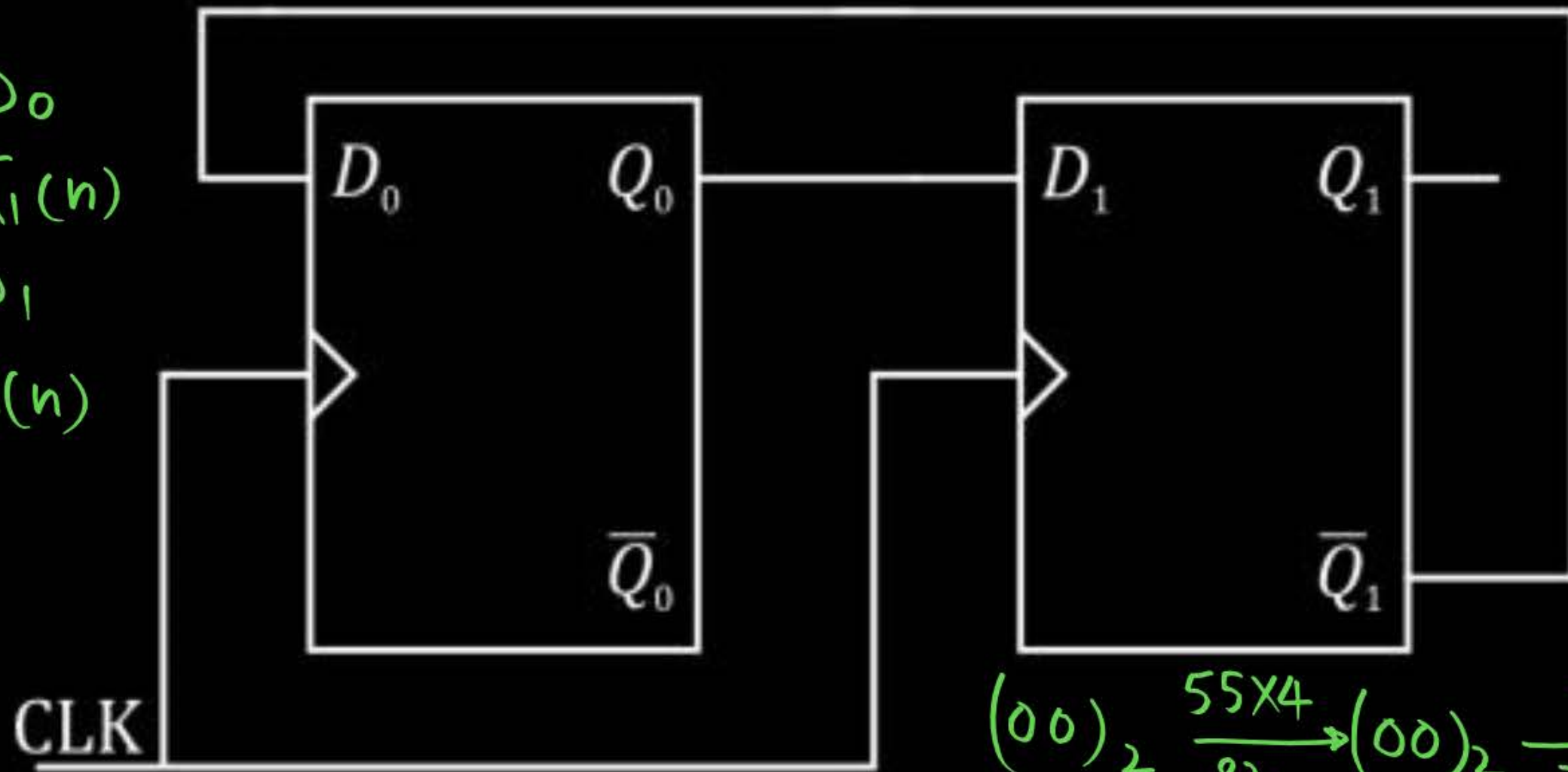
$$Q_1(n+1) = A\bar{Q}_1(n) + A Q_1(n) = A$$

# [ Question ]

A sequential circuit is as given below:  $(0-2-3-1)$

$$Q_0(n+1) = D_0 = \overline{Q_1}(n)$$

$$Q_1(n+1) = D_1 = Q_0(n)$$



freq/4

$Q_0 \quad Q_1$

|   |   |   |
|---|---|---|
|   | 0 | 0 |
| ↑ | 1 | 0 |
| ↑ | 1 | 1 |
| ↑ | 0 | 1 |
| ↑ | 0 | 0 |

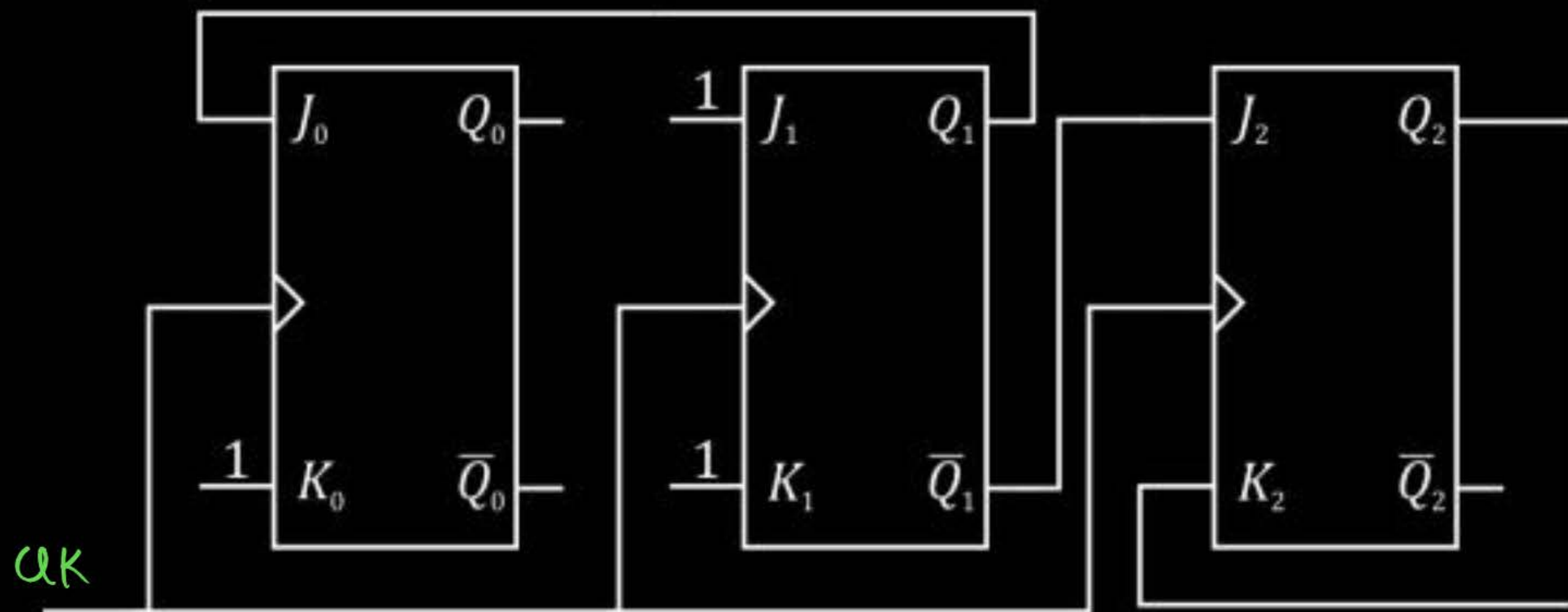
$$(00)_2 \xrightarrow[220]{55 \times 4} (00)_2 \xrightarrow[20K]{20K} (11)_2$$

Initial state of the counter is  $Q_0Q_1 = (00)_2$ . What will the state of counter after application of 222 clock pulses  $(3)_{10}$ .



# [ Question ]

A sequential circuit is as given below :



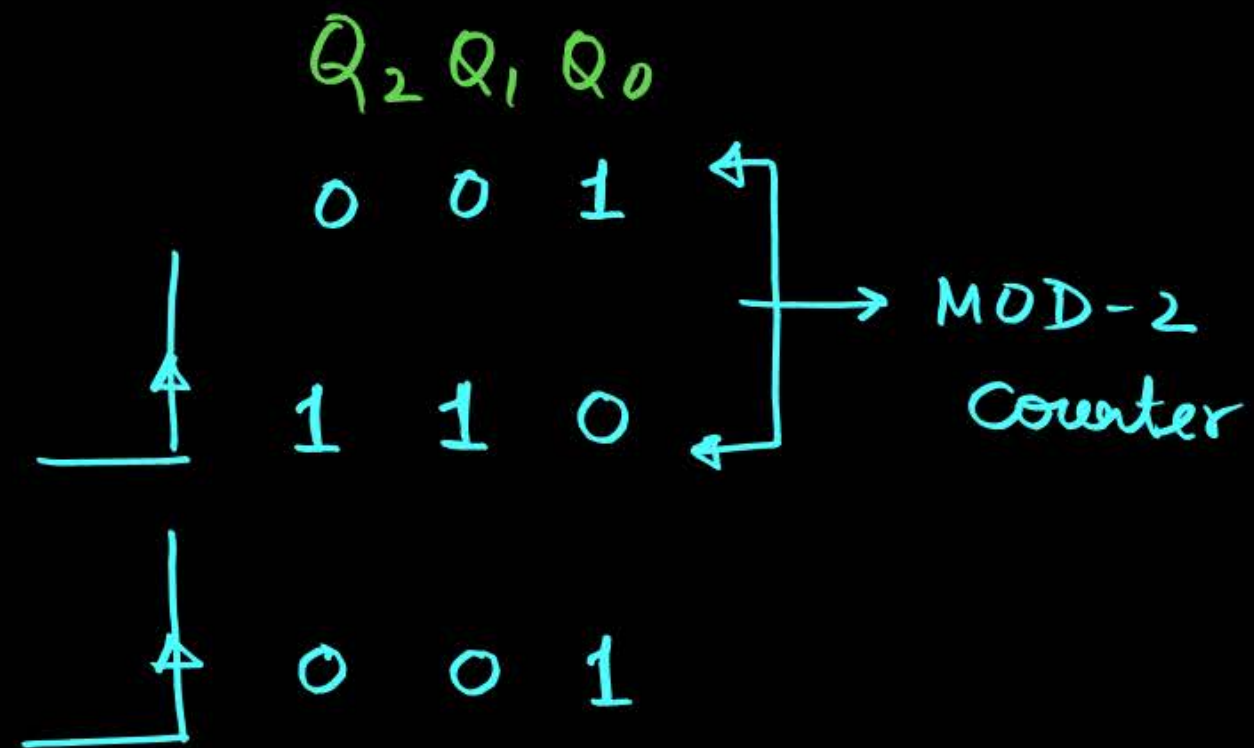
If initial state of counter is  $Q_2Q_1Q_0 = (001)_2$ , then after 8<sup>th</sup> clock pulse, counter will be in state ( **1** )<sub>10</sub>.

$$Q_2(n+1) = J_2 \bar{Q}_2(n) + \bar{K}_2 Q_2(n)$$

$$= \bar{Q}_1(n) \cdot \bar{Q}_2(n) = \overline{Q_1(n) + Q_2(n)}$$

$$Q_1(n+1) = \bar{Q}_1(n)$$

$$Q_0(n+1) = Q_1(n) \cdot \bar{Q}_0(n)$$



$$(001)_2 \xrightarrow{4 \times 2} (001)_2$$

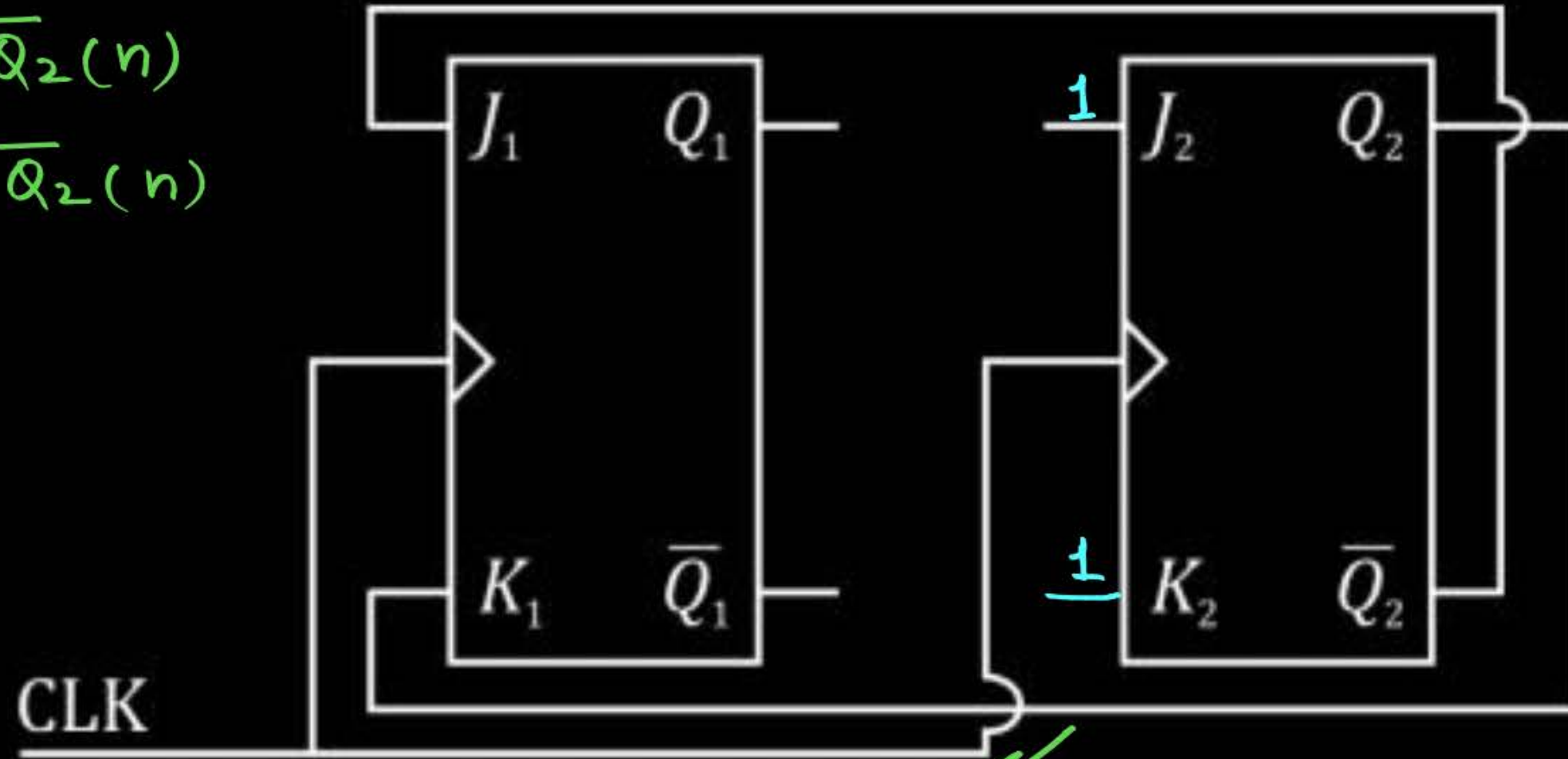


# [ Question ]

The output of the two FFs,  $Q_1$  and  $Q_2$  are initially at '0'. The sequence generated at  $Q_1$  after application of clock pulses :

$$Q_1(n+1) = \overline{Q_2}(n)$$

$$Q_2(n+1) = \overline{Q_1}(n)$$



|   | $Q_1$ | $Q_2$ |
|---|-------|-------|
|   | 0     | 0     |
| ↑ | 1     | 1     |
| ↑ | 0     | 0     |
| ↑ | 1     | 1     |

(a) 01110.....

(c) 00110.....

✓ (b) 01010.....

(d) 01100.....

# [ Synchronous Series Carry Counter ]



4-bit synchronous series carry counter :

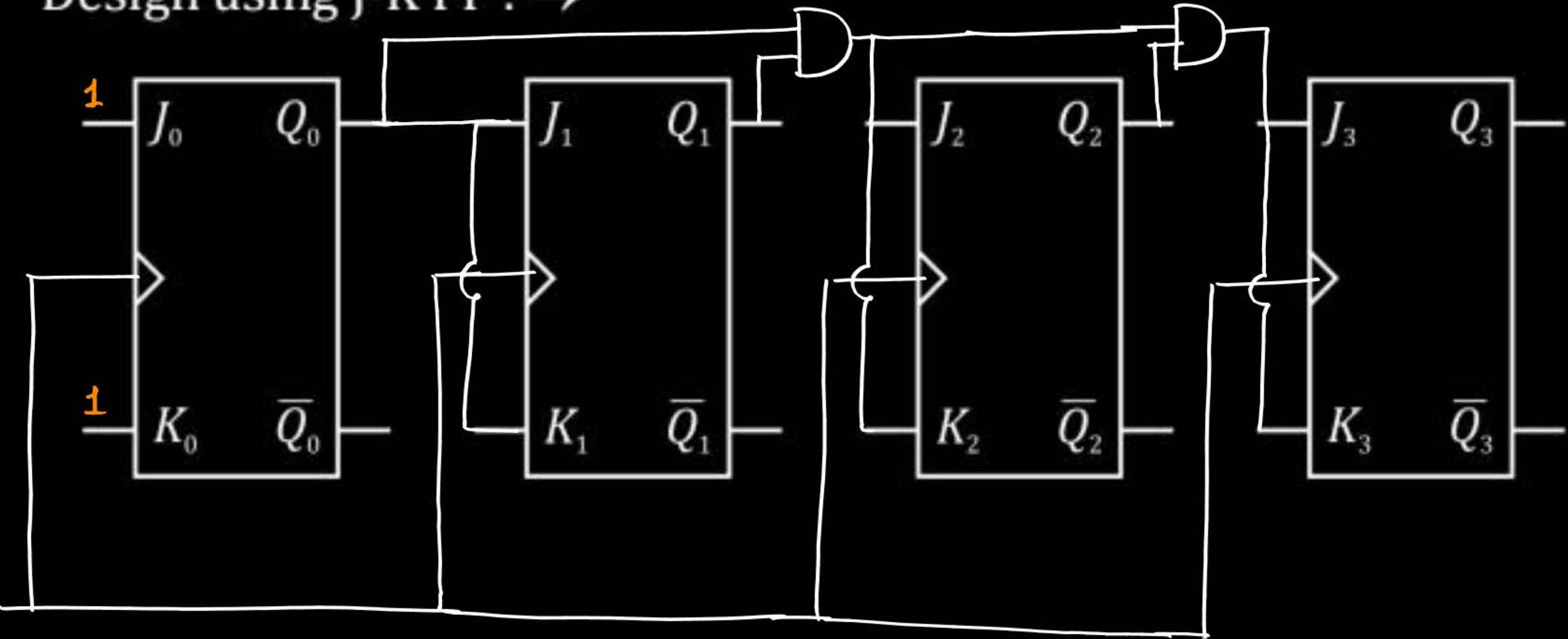
Let's analyze the sequence and design it :

| $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|-------|-------|-------|-------|
| 0     | 0     | 0     | 0     |
| 0     | 0     | 0     | 1     |
| 0     | 0     | 1     | 0     |
| 0     | 0     | 1     | 1     |
| 0     | 1     | 0     | 0     |
| 0     | 1     | 1     | 0     |
| 0     | 1     | 1     | 1     |
| 1     | 0     | 0     | 0     |
| 1     | 0     | 0     | 1     |
| 1     | 0     | 1     | 0     |

| $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|-------|-------|-------|-------|
| 1     | 0     | 1     | 1     |
| 1     | 1     | 0     | 0     |
| 1     | 1     | 0     | 1     |
| 1     | 1     | 1     | 0     |
| 1     | 1     | 1     | 1     |
| 0     | 0     | 0     | 0     |
|       |       |       |       |
|       |       |       |       |
|       |       |       |       |
|       |       |       |       |
|       |       |       |       |



Design using J-K FF :  $\rightarrow$

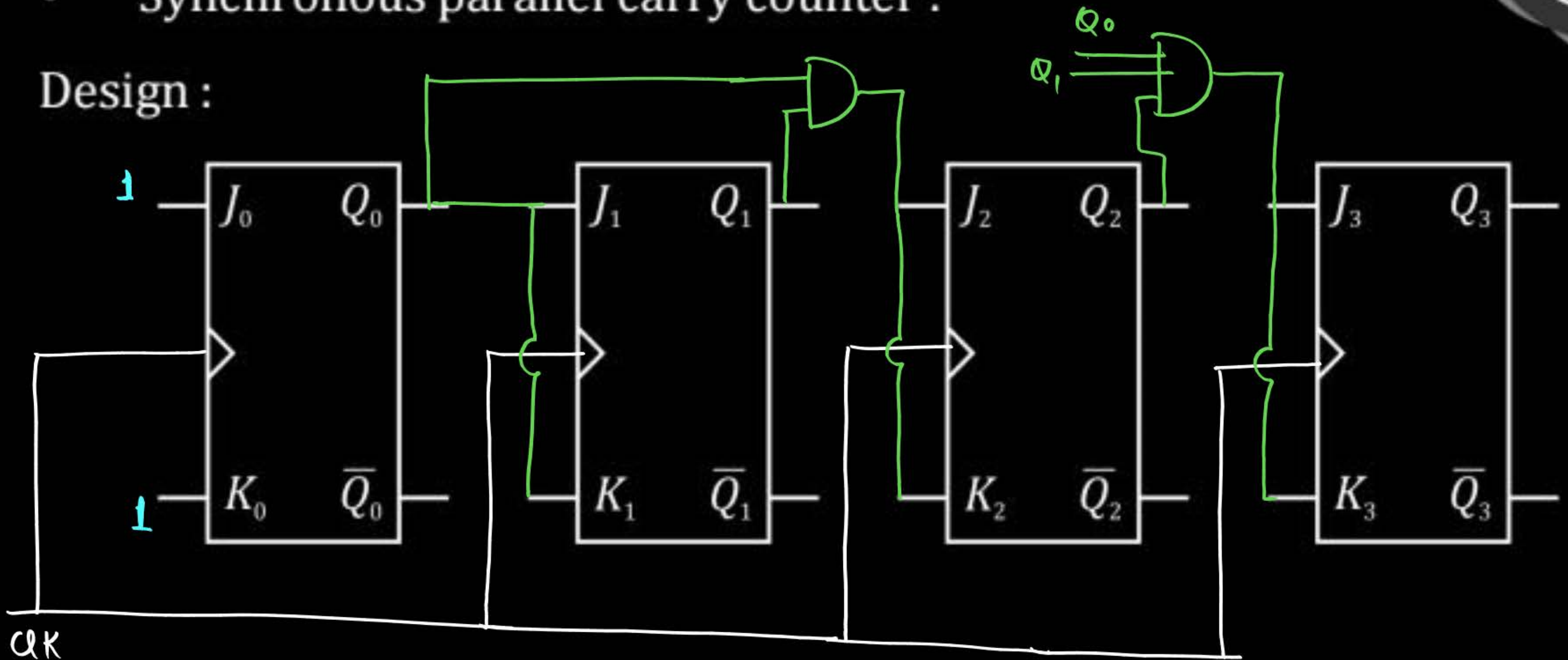


No. of AND gate require is :

→  $(n-2)$  AND gates  $[2-C/P]$  for  $n$  bit synchronous series Carry Counter.

- Synchronous parallel carry counter :

Design :





- $(n-2)$  AND gates will be required for  $n$ -bit synchronous parallel carry counter.
- Highest I/P AND gate  $\rightarrow (n-1)$



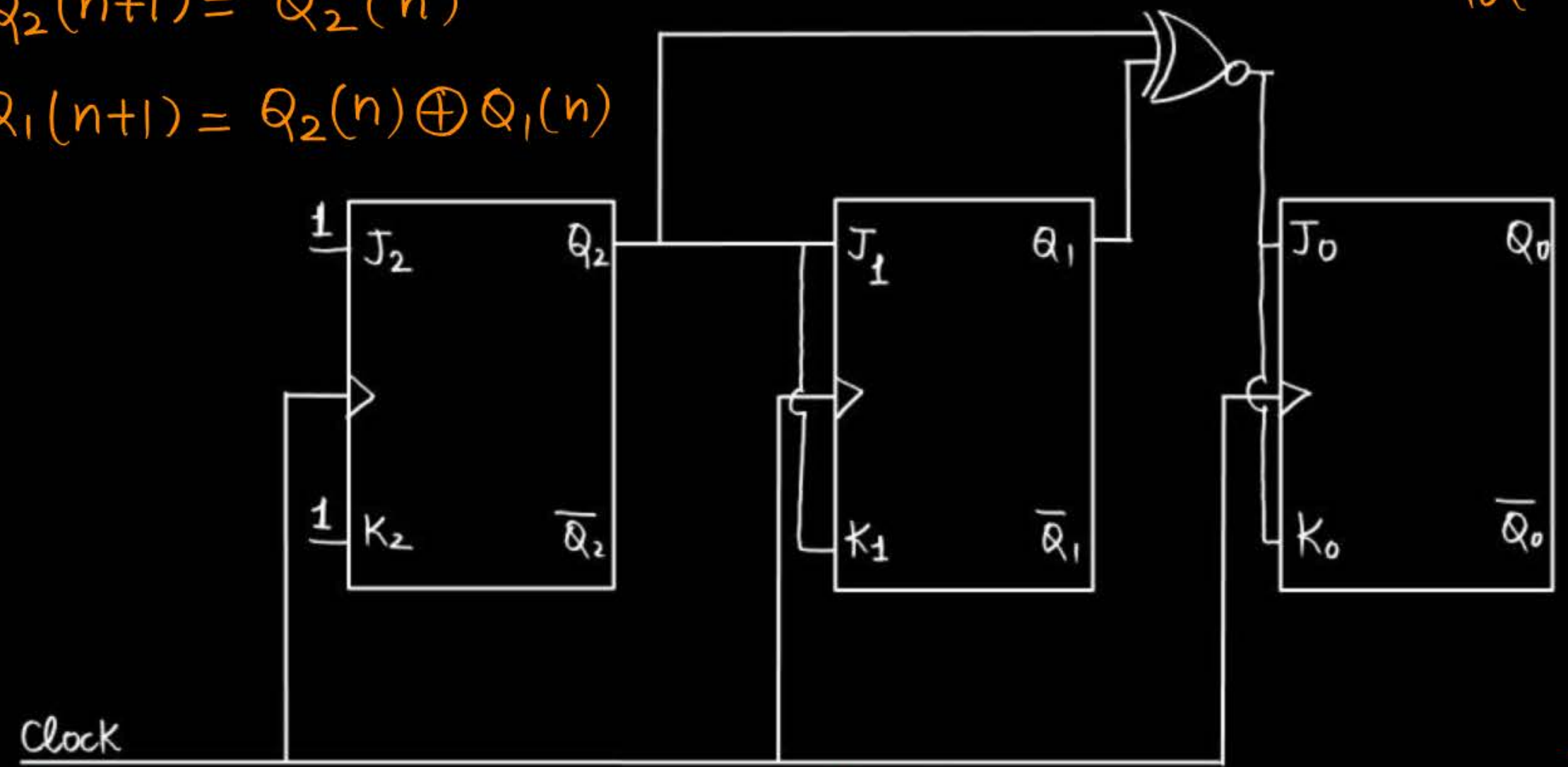


#Q. A sequential CKt is as given below:

$Q_2(n+1) = \overline{Q_2}(n)$

$Q_1(n+1) = Q_2(n) \oplus Q_1(n)$

$Q_0(n+1) = Q_2(n) \odot Q_1(n) \oplus Q_0(n)$   
 $= \overline{Q_2(n) \oplus Q_1(n) \oplus Q_0(n)}$



|   | $Q_2$ | $Q_1$ | $Q_0$ |
|---|-------|-------|-------|
|   | 0     | 0     | 0     |
| ↑ | 1     | 0     | 1     |
| ↑ | 0     | 1     | 1     |
| ↑ | 1     | 1     | 1     |

MOD-4

The ckt starts with  $Q_2Q_1Q_0 = (000)_2$  then next three states of the counter will be:

- ~~a. 101, 111, 011~~ ~~b. 101, 010, 111~~ c. 101, 011, 111 ~~d. 111, 011, 101~~

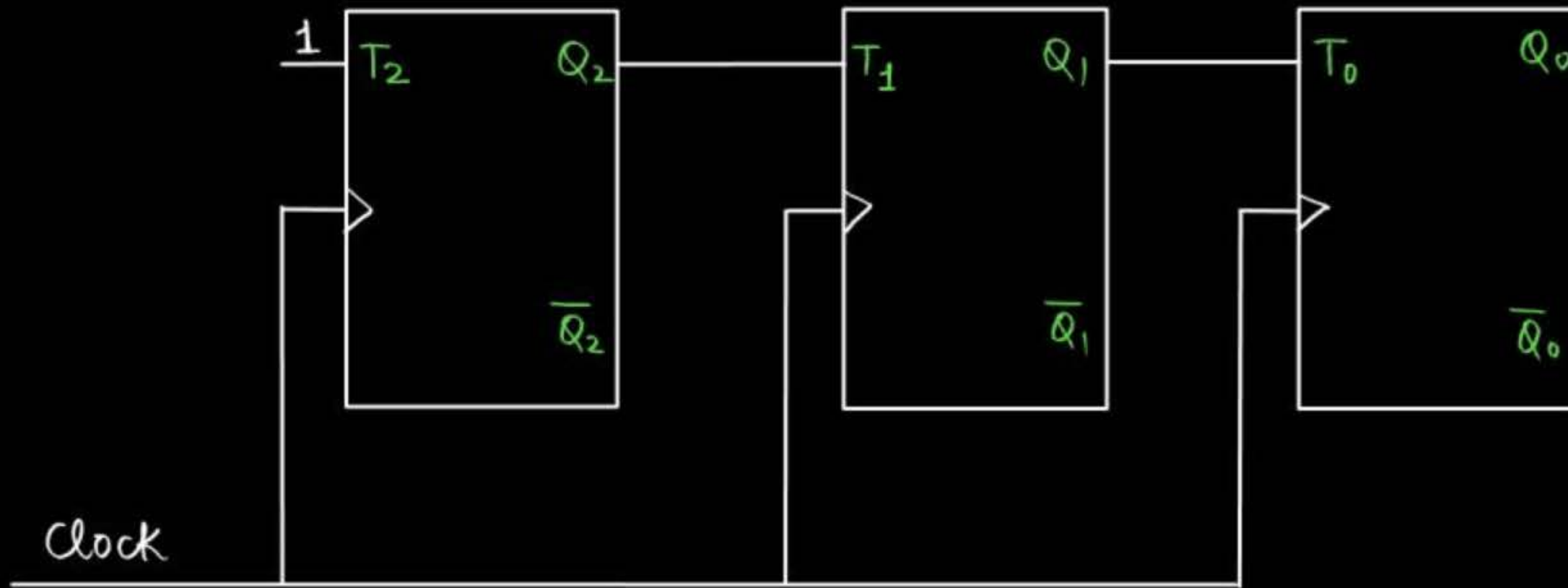
|   |   |   |   |
|---|---|---|---|
| ↑ | 0 | 0 | 0 |
|---|---|---|---|



#Q. A sequential CKt is as given below:

$$Q_2(n+1) = \overline{Q_2}(n) \quad Q_1(n+1) = T_1 \oplus Q_1(n) = Q_2(n) \oplus Q_1(n)$$

$$Q_0(n+1) = T_0 \oplus Q_0(n) = Q_1(n) \oplus Q_0(n)$$



|           | $Q_2$ | $Q_1$ | $Q_0$ |
|-----------|-------|-------|-------|
| Initial   | 1     | 0     | 0     |
| 1st clock | 0     | 1     | 0     |
| 2nd clock | 1     | 1     | 1     |
| 3rd clock | 0     | 0     | 0     |

The CKt starts with  $Q_2Q_1Q_0 = (100)_2$  then state of the counter after 3<sup>rd</sup> clock will be

- ~~a.  $(010)_2$~~  ~~b.  $(111)_2$~~  ~~c.  $(100)_2$~~  ☒ d.  $(000)_2$



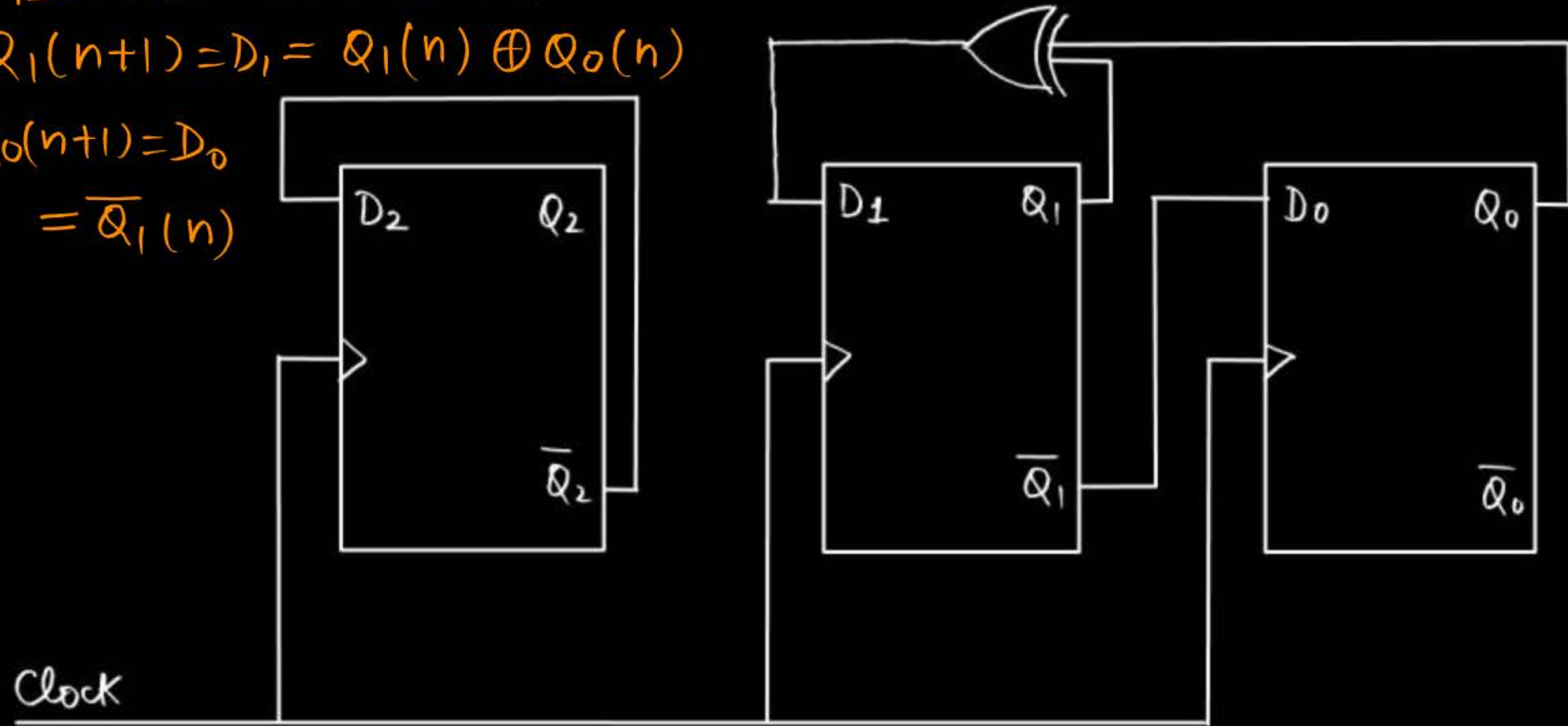


#Q. A sequential CKt is as given below:

$$Q_2(n+1) = D_2 = \overline{Q_2}(n)$$

$$Q_1(n+1) = D_1 = Q_1(n) \oplus Q_0(n)$$

$$Q_0(n+1) = D_0 = \overline{Q_1}(n)$$



|   | $Q_2$ | $Q_1$ | $Q_0$ |
|---|-------|-------|-------|
|   | 1     | 0     | 1     |
| ↑ | 0     | 1     | 1     |
| ↑ | 1     | 0     | 0     |
| ↑ | 0     | 0     | 1     |
| ↑ | 1     | 1     | 1     |

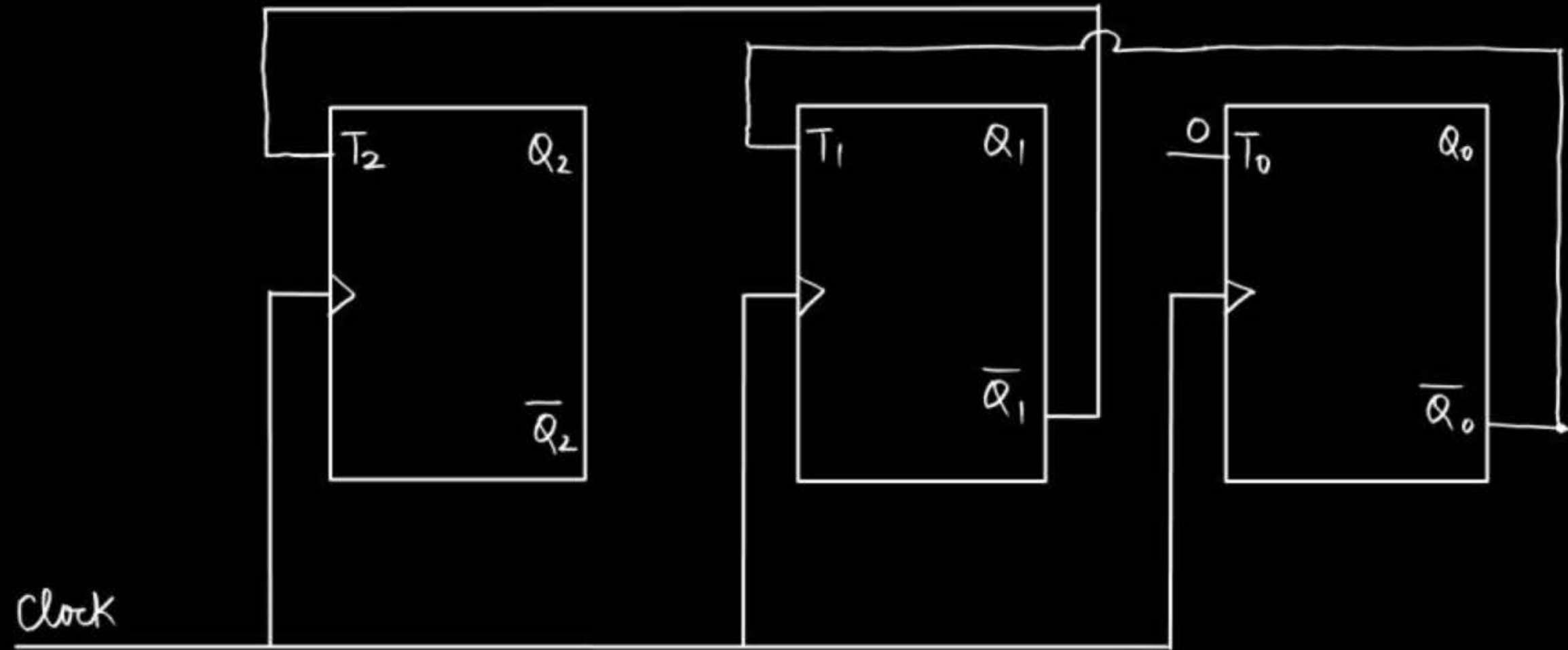
a. If starting state of the counter is  $Q_2 Q_1 Q_0 = (101)_2$  then what is the state of the counter after 4<sup>th</sup> clock edge (7)<sub>10</sub>.

b. MOD no. of the counter 6.

|   |   |   |   |
|---|---|---|---|
| ↑ | 0 | 0 | 0 |
| ↑ | 1 | 0 | 1 |



#Q. A sequential CKt is as given below:

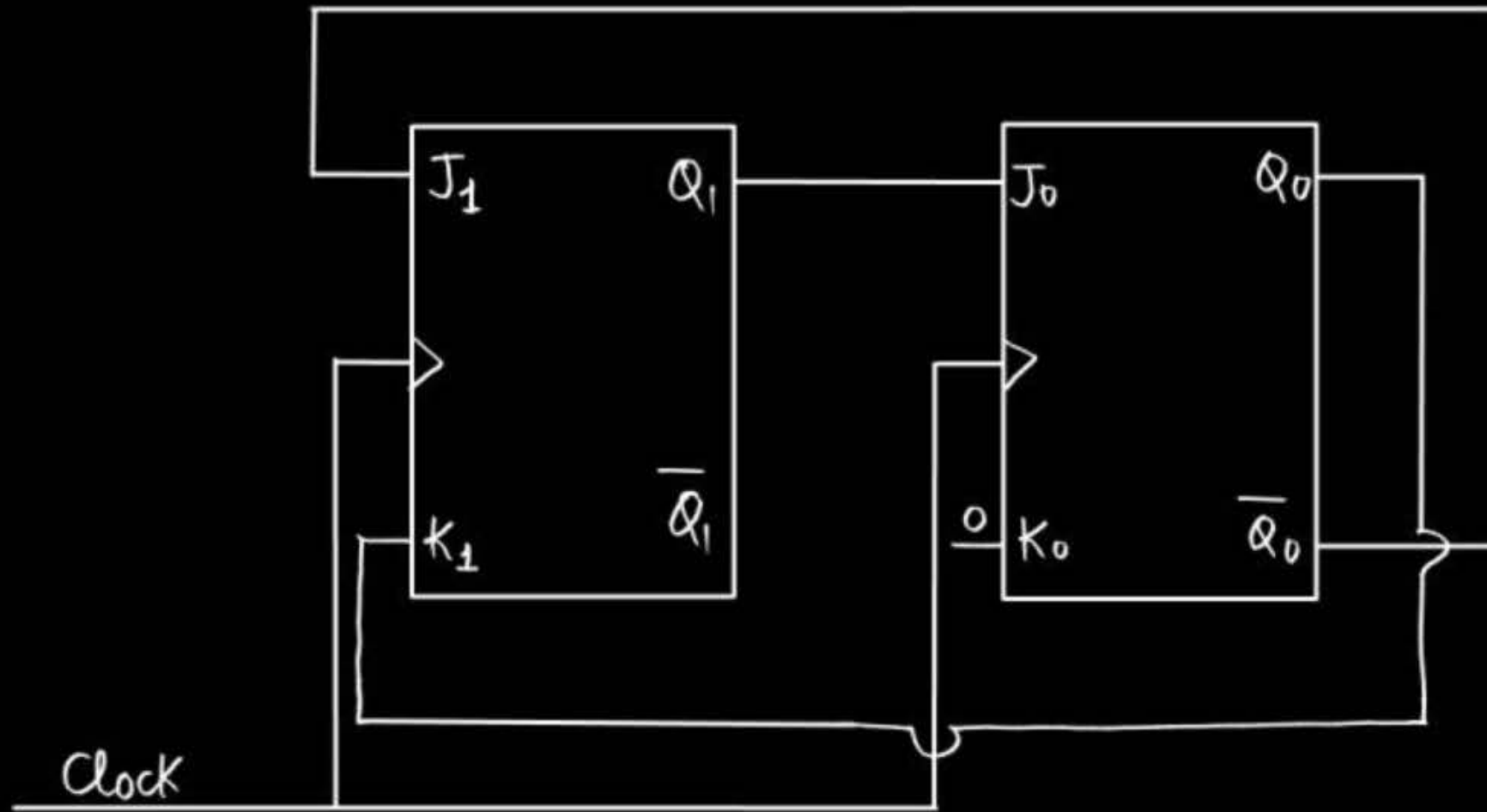


H.W.

- a. If the input clock frequency is  $10 \text{ MHz}$ , then o/p frequency at o/p  $Q_1$  will be \_\_\_\_\_  $\text{MHz}$ .
- b. If counter starts with  $(110)_2$  then summation of next three states will be \_\_\_\_\_.



#Q. A sequential CKt is as given below:



H.W.

Given Counter started with  $Q_1 Q_0 = (00)_2$  then next four state of the counter will be:  
a. 10, 11, 01, 00    b. 10, 11, 00, 10    c. 10, 11, 01, 01    d. 01, 11, 10, 10





## Topic : 2 Min Summary

→ Synch. Counter.

**Thank you**

**GW**  
*Soldiers !*

